

13 june

In [6]:

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

In [7]:

```
heart_data=pd.read_csv(r"C:\Users\kavya\Downloads\heart.csv")
```

In [8]:

```
heart_data
```

Out[8]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	0
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	0
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	0
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	0
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	0

303 rows × 14 columns

In [12]:

```
heart_data.shape
```

Out[12]:

```
(303, 14)
```

In [14]:

```
heart_data.head()
```

Out[14]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

In [16]:

```
heart_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   age         303 non-null    int64
 1   sex         303 non-null    int64
 2   cp          303 non-null    int64
 3   trestbps    303 non-null    int64
 4   chol        303 non-null    int64
 5   fbs         303 non-null    int64
 6   restecg     303 non-null    int64
 7   thalach     303 non-null    int64
 8   exang       303 non-null    int64
 9   oldpeak     303 non-null    float64
10   slope       303 non-null    int64
11   ca          303 non-null    int64
12   thal        303 non-null    int64
13   target      303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [18]:

```
heart_data.dtypes
```

Out[18]:

```
age         int64
sex         int64
cp          int64
trestbps    int64
chol        int64
fbs         int64
restecg     int64
thalach     int64
exang       int64
oldpeak     float64
slope       int64
ca          int64
thal        int64
target      int64
dtype: object
```

In [20]:

```
heart_data.describe()
```

Out[20]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000

In [22]:

```
heart_data.columns
```

Out[22]:

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

In [24]:

```
heart_data['target'].nunique()
```

Out[24]:

```
2
```

In [26]:

```
heart_data['target'].value_counts()
```

Out[26]:

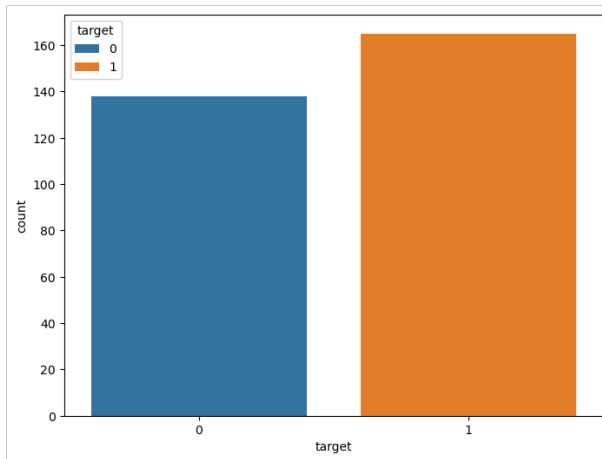
```
target
1    165
0    138
Name: count, dtype: int64
```

In [28]:

```
f,ax=plt.subplots(figsize=(8,6))
sns.countplot(x='target',data=heart_data,hue='target')
```

Out[28]:

<Axes: xlabel='target', ylabel='count'>



In [30]:

```
heart_data.groupby('sex')['target'].value_counts()
```

Out[30]:

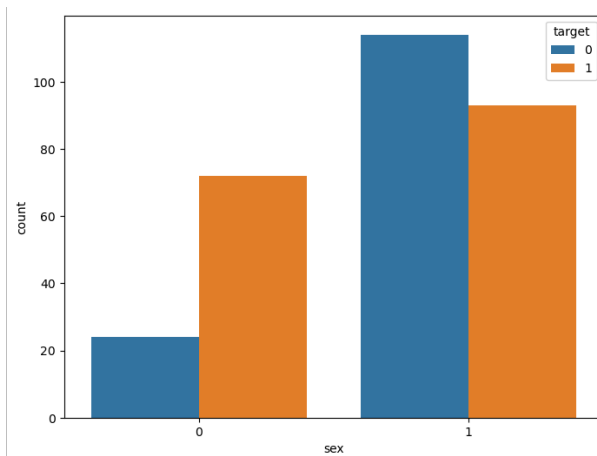
```
sex  target
0    1      72
    0      24
1    0     114
    1      93
Name: count, dtype: int64
```

In [32]:

```
f,ax=plt.subplots(figsize=(8,6))
sns.countplot(data=heart_data,x='sex',hue='target')
```

Out[32]:

<Axes: xlabel='sex', ylabel='count'>

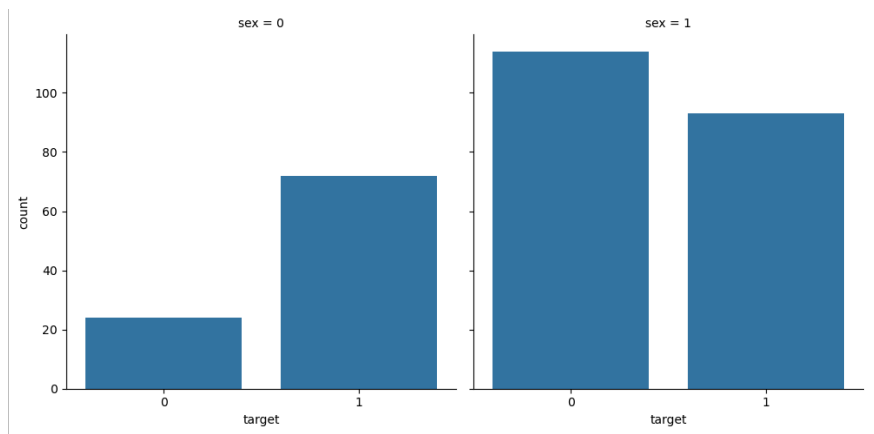


In [34]:

```
sns.catplot(data=heart_data,x="target",col="sex",kind="count",height=5,aspect=1)
```

Out[34]:

<seaborn.axisgrid.FacetGrid at 0x19ebca661e0>

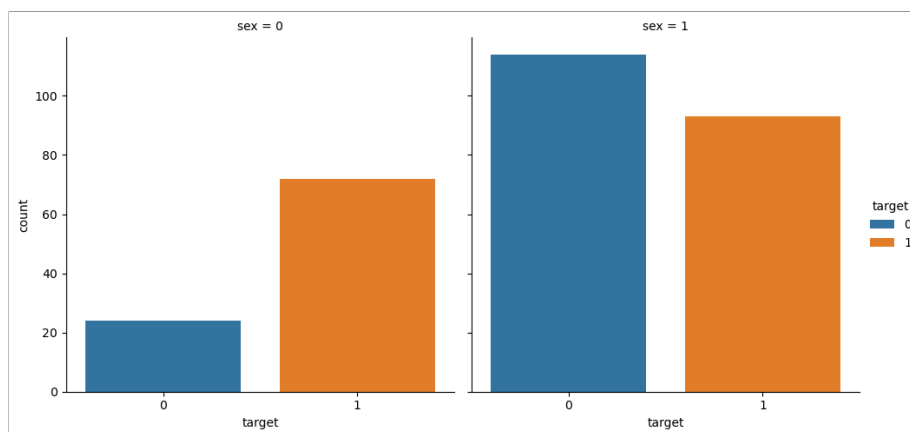


In [35]:

```
sns.catplot(x="target", col="sex", data=heart_data, kind="count", height=5, aspect=1, hue='target')
```

Out[35]:

```
<seaborn.axisgrid.FacetGrid at 0x19ebca64ec0>
```

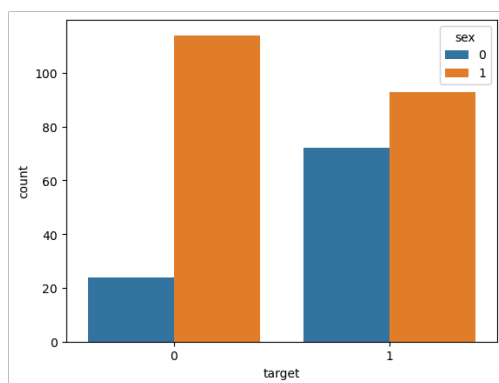


In [37]:

```
sns.countplot(data=heart_data, x='target', hue='sex')
```

Out[37]:

```
<Axes: xlabel='target', ylabel='count'>
```

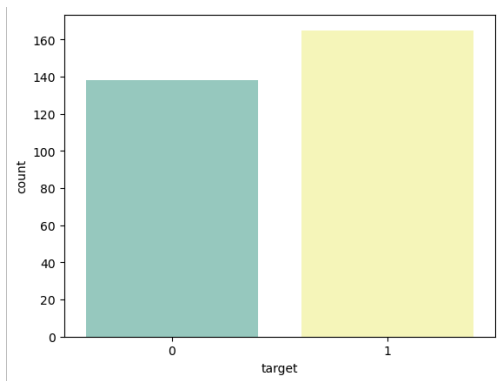


In [40]:

```
sns.countplot(data=heart_data, x='target', palette='Set3')
```

Out[40]:

```
<Axes: xlabel='target', ylabel='count'>
```

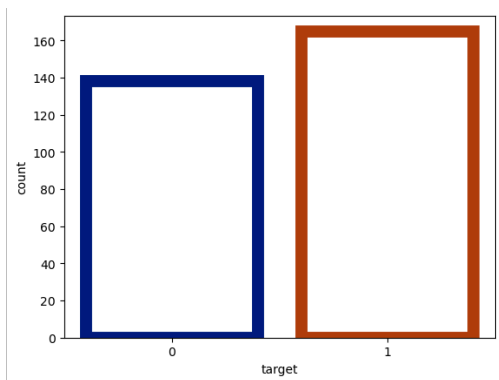


In [42]:

```
sns.countplot(data=heart_data,x="target",facecolor=(0,0,0,0),linewidth=10,edgecolor=sns.color_palette("dark",3))
```

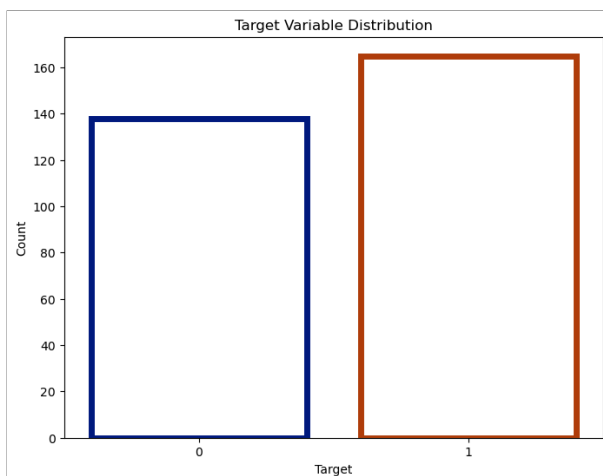
Out[42]:

<Axes: xlabel='target', ylabel='count'>



In [44]:

```
plt.figure(figsize=(8, 6))
sns.countplot(
    data=heart_data,
    x="target",
    facecolor=(0, 0, 0, 0),
    linewidth=5,
    edgecolor=sns.color_palette("dark", 2)
)
plt.title("Target Variable Distribution")
plt.xlabel("Target")
plt.ylabel("Count")
plt.show()
```

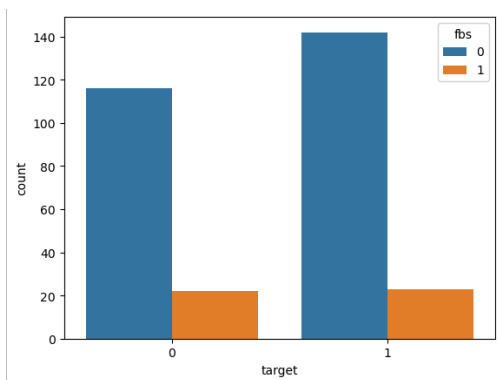


In [46]:

```
sns.countplot(data=heart_data,x="target",hue="fbs")
```

Out[46]:

<Axes: xlabel='target', ylabel='count'>

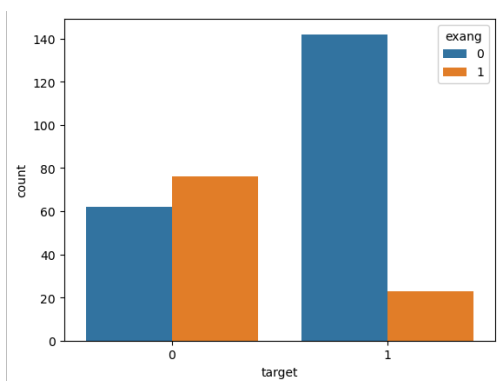


In [48]:

```
sns.countplot(data=heart_data, x="target", hue="exang")
```

Out[48]:

```
<Axes: xlabel='target', ylabel='count'>
```



In [50]:

```
correlation=heart_data.corr()
```

In [52]:

```
correlation['target'].sort_values(ascending=False)
```

Out[52]:

```
target      1.000000
cp          0.433798
thalach     0.421741
slope       0.345877
restecg     0.137230
fbs        -0.028046
chol       -0.085239
trestbps   -0.144931
age        -0.225439
sex        -0.280937
thal       -0.344029
ca         -0.391724
oldpeak    -0.430696
exang      -0.436757
Name: target, dtype: float64
```

In [54]:

```
correlation['target'].sort_values(ascending=True)
```

Out[54]:

```
exang       -0.436757
oldpeak     -0.430696
ca          -0.391724
thal        -0.344029
sex         -0.280937
age         -0.225439
trestbps    -0.144931
chol        -0.085239
fbs         -0.028046
restecg     0.137230
slope       0.345877
thalach     0.421741
cp          0.433798
target      1.000000
Name: target, dtype: float64
```

In [56]:

```
heart_data["cp"].nunique()
```

```
Out[56]:
```

```
4
```

```
In [58]:
```

```
heart_data["cp"].value_counts()
```

```
Out[58]:
```

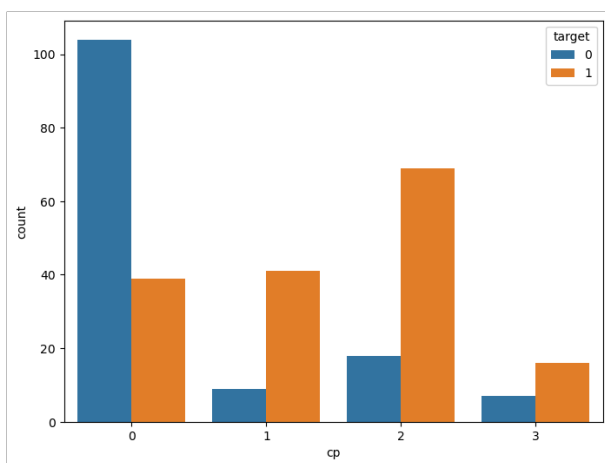
```
cp
0    143
2     87
1     50
3     23
Name: count, dtype: int64
```

```
In [60]:
```

```
f,ax=plt.subplots(figsize=(8,6))
sns.countplot(data=heart_data,x="cp",hue='target')
```

```
Out[60]:
```

```
<Axes: xlabel='cp', ylabel='count'>
```



```
In [62]:
```

```
heart_data.groupby('cp')['target'].value_counts()
```

```
Out[62]:
```

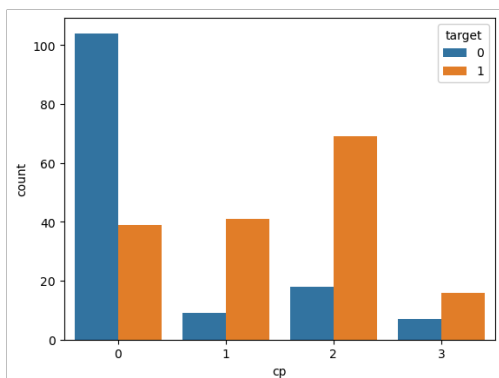
```
cp  target
0    0      104
    1       39
1    1       41
    0        9
2    1       69
    0       18
3    1       16
    0        7
Name: count, dtype: int64
```

```
In [64]:
```

```
sns.countplot(data=heart_data,x="cp",hue='target')
```

```
Out[64]:
```

```
<Axes: xlabel='cp', ylabel='count'>
```

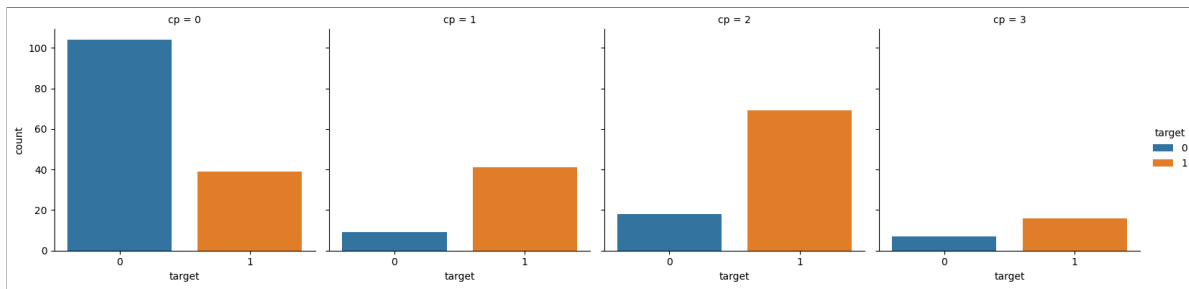


```
In [66]:
```

```
sns.catplot(data=heart_data,x="target",col="cp",kind="count",height=4,aspect=1,hue="target")
```

Out[66]:

<seaborn.axisgrid.FacetGrid at 0x19ebcfc4440>



In [67]:

heart_data["thalach"].nunique()

Out[67]:

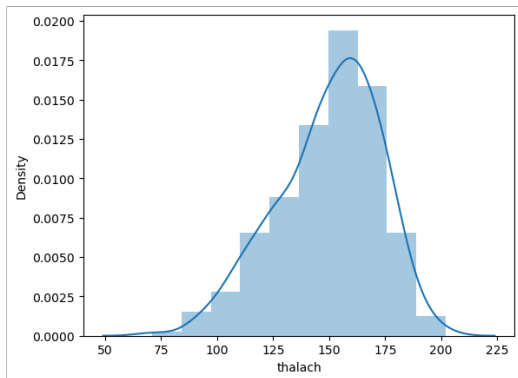
91

In [70]:

sns.distplot(heart_data["thalach"], bins=10)

Out[70]:

<Axes: xlabel='thalach', ylabel='Density'>

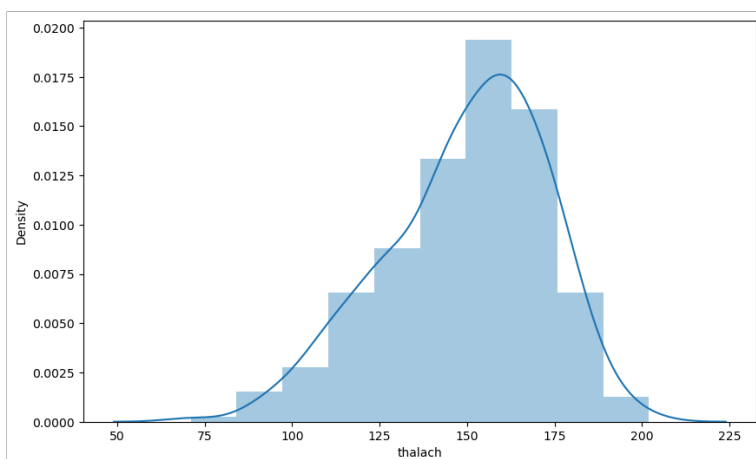


In [72]:

```
plt.figure(figsize=(10,6))
sns.distplot(heart_data["thalach"], bins=10)
```

Out[72]:

<Axes: xlabel='thalach', ylabel='Density'>

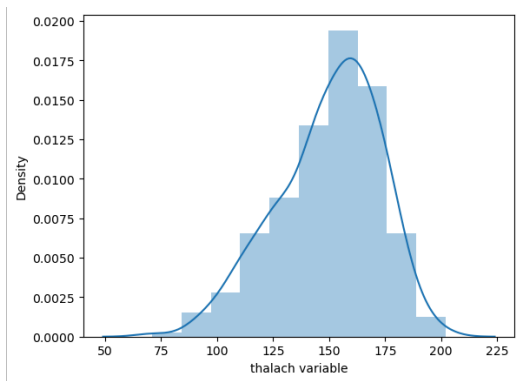


In [74]:

sns.distplot(pd.Series(heart_data["thalach"], name="thalach variable"), bins=10)

Out[74]:

<Axes: xlabel='thalach variable', ylabel='Density'>

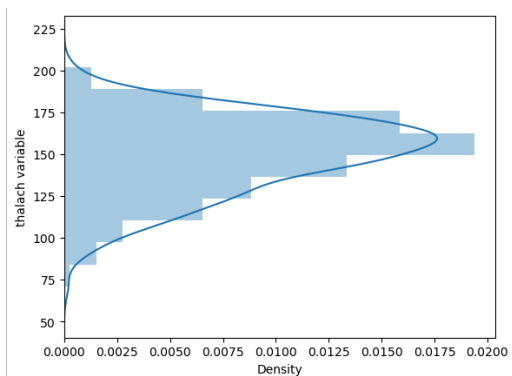


In [76]:

```
sns.distplot(pd.Series(heart_data["thalach"],name="thalach variable"),bins=10,vertical=True)
```

Out[76]:

```
<Axes: xlabel='Density', ylabel='thalach variable'>
```

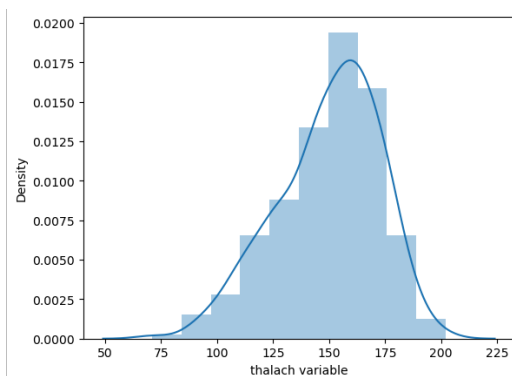


In [78]:

```
sns.distplot(pd.Series(heart_data["thalach"],name="thalach variable"),bins=10,vertical=False)
```

Out[78]:

```
<Axes: xlabel='thalach variable', ylabel='Density'>
```

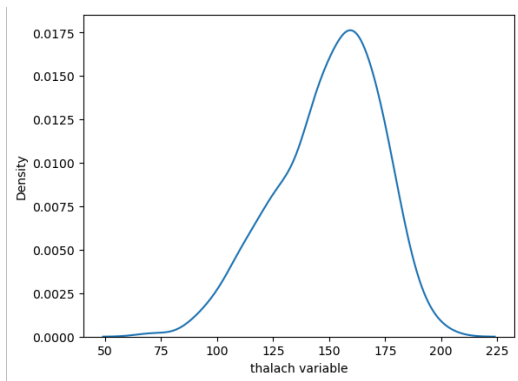


In [80]:

```
sns.kdeplot(pd.Series(heart_data["thalach"],name="thalach variable"))
```

Out[80]:

```
<Axes: xlabel='thalach variable', ylabel='Density'>
```

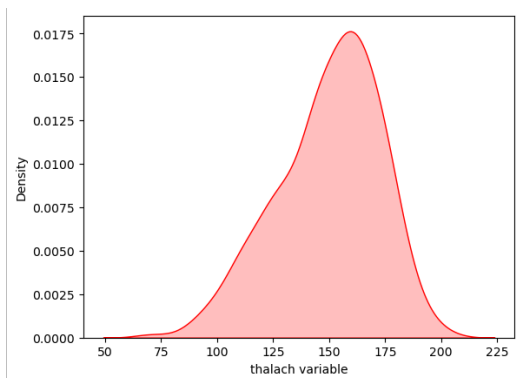


In [82]:

```
sns.kdeplot(pd.Series(heart_data["thalach"],name="thalach variable"),shade=True,color='r')
```

Out[82]:

<Axes: xlabel='thalach variable', ylabel='Density'>

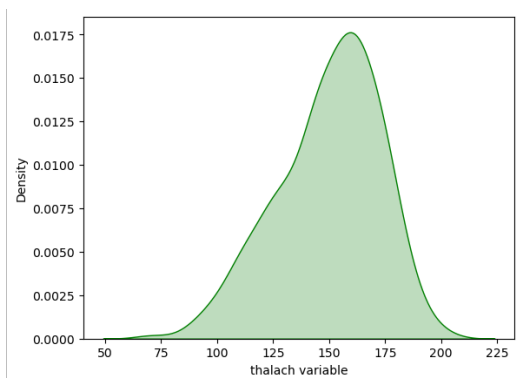


In [84]:

```
sns.kdeplot(pd.Series(heart_data["thalach"],name="thalach variable"),shade=True,color='g')
```

Out[84]:

<Axes: xlabel='thalach variable', ylabel='Density'>

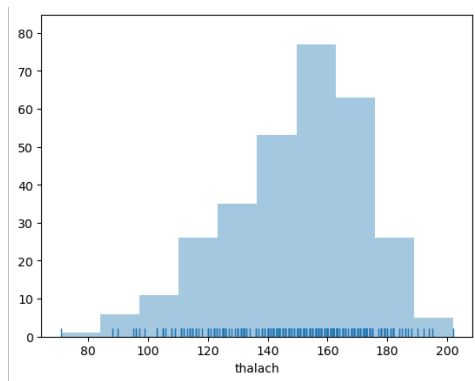


In [86]:

```
sns.distplot(heart_data["thalach"],kde=False,rug=True,bins=10)
```

Out[86]:

<Axes: xlabel='thalach'>

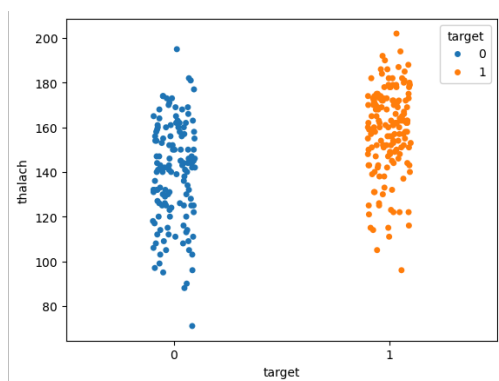


In [88]:

```
sns.stripplot(data=heart_data,x="target",y="thalach",hue="target")
```

Out[88]:

<Axes: xlabel='target', ylabel='thalach'>

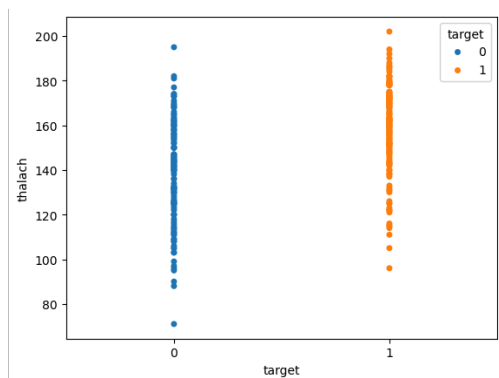


In [90]:

```
sns.stripplot(data=heart_data,x="target",y="thalach",jitter=0.00001,hue="target")
```

Out[90]:

<Axes: xlabel='target', ylabel='thalach'>

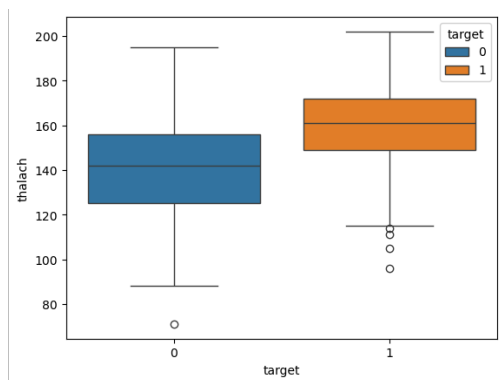


In [92]:

```
sns.boxplot(data=heart_data,x="target",y="thalach",hue="target")
```

Out[92]:

<Axes: xlabel='target', ylabel='thalach'>

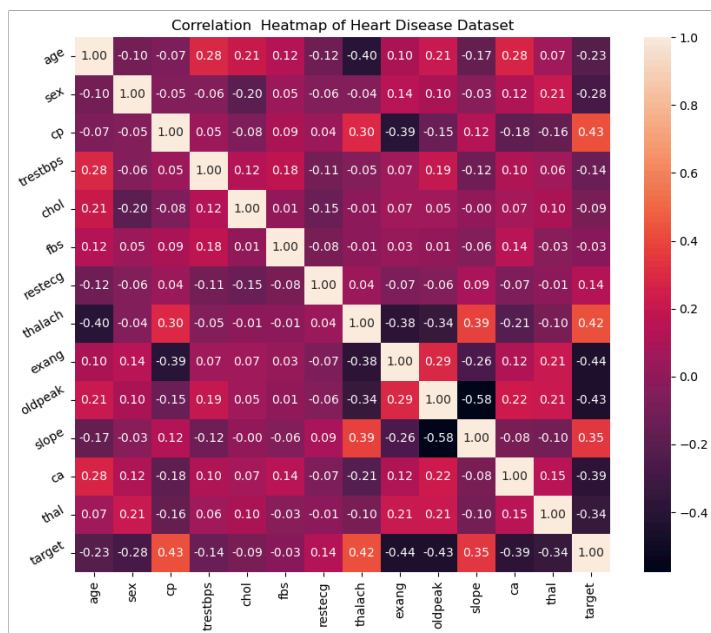


In [94]:

```
plt.figure(figsize=(10,8))
plt.title("Correlation Heatmap of Heart Disease Dataset")
a=sns.heatmap(correlation,square=True,annot=True,fmt='.2f',linecolor='white')
a.set_xticklabels(a.get_xticklabels(),rotation=90)
a.set_yticklabels(a.get_yticklabels(),rotation=30)
```

Out[94]:

```
[Text(0, 0.5, 'age'),
 Text(0, 1.5, 'sex'),
 Text(0, 2.5, 'cp'),
 Text(0, 3.5, 'trestbps'),
 Text(0, 4.5, 'chol'),
 Text(0, 5.5, 'fbs'),
 Text(0, 6.5, 'restecg'),
 Text(0, 7.5, 'thalach'),
 Text(0, 8.5, 'exang'),
 Text(0, 9.5, 'oldpeak'),
 Text(0, 10.5, 'slope'),
 Text(0, 11.5, 'ca'),
 Text(0, 12.5, 'thal'),
 Text(0, 13.5, 'target')]
```

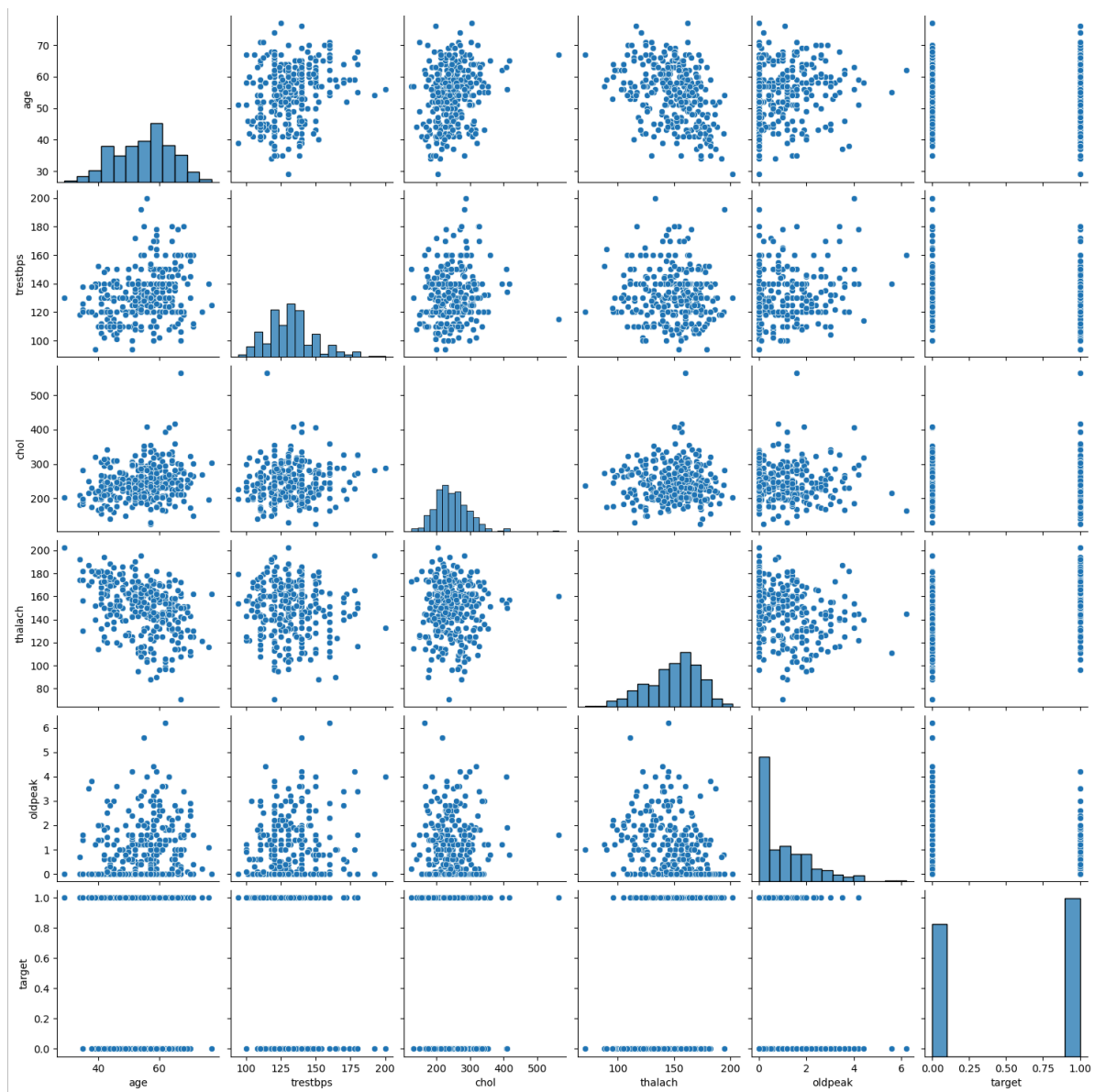


In [96]:

```
num_var=["age","trestbps","chol","thalach","oldpeak","target"]
sns.pairplot(heart_data[num_var],kind='scatter',diag_kind='hist')
```

Out[96]:

```
<seaborn.axisgrid.PairGrid at 0x19ec2093d10>
```

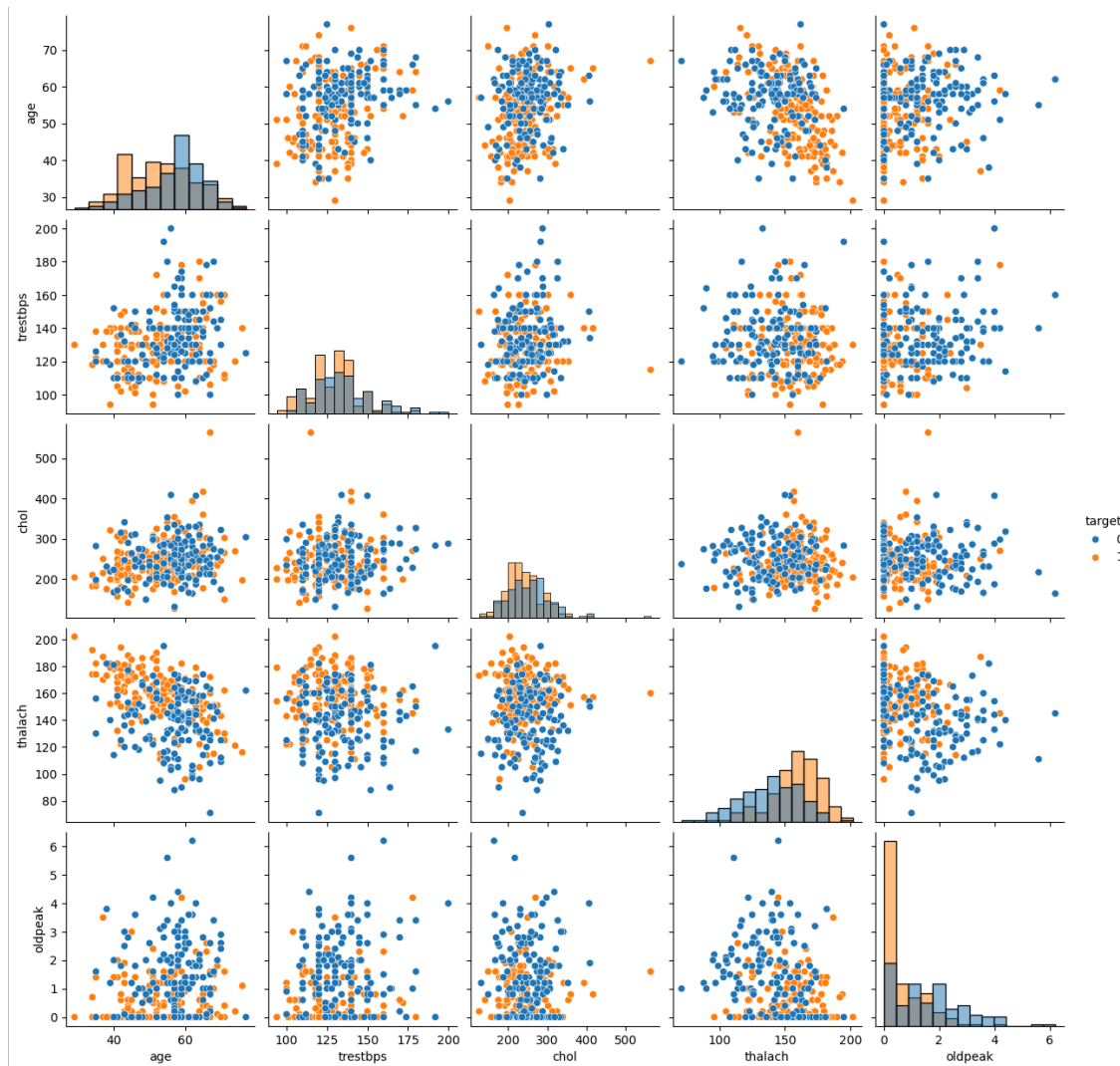


In [99]:

```
sns.pairplot(heart_data[num_var], kind='scatter', diag_kind='hist', hue='target')
```

Out[99]:

```
<seaborn.axisgrid.PairGrid at 0x19ec59f24b0>
```

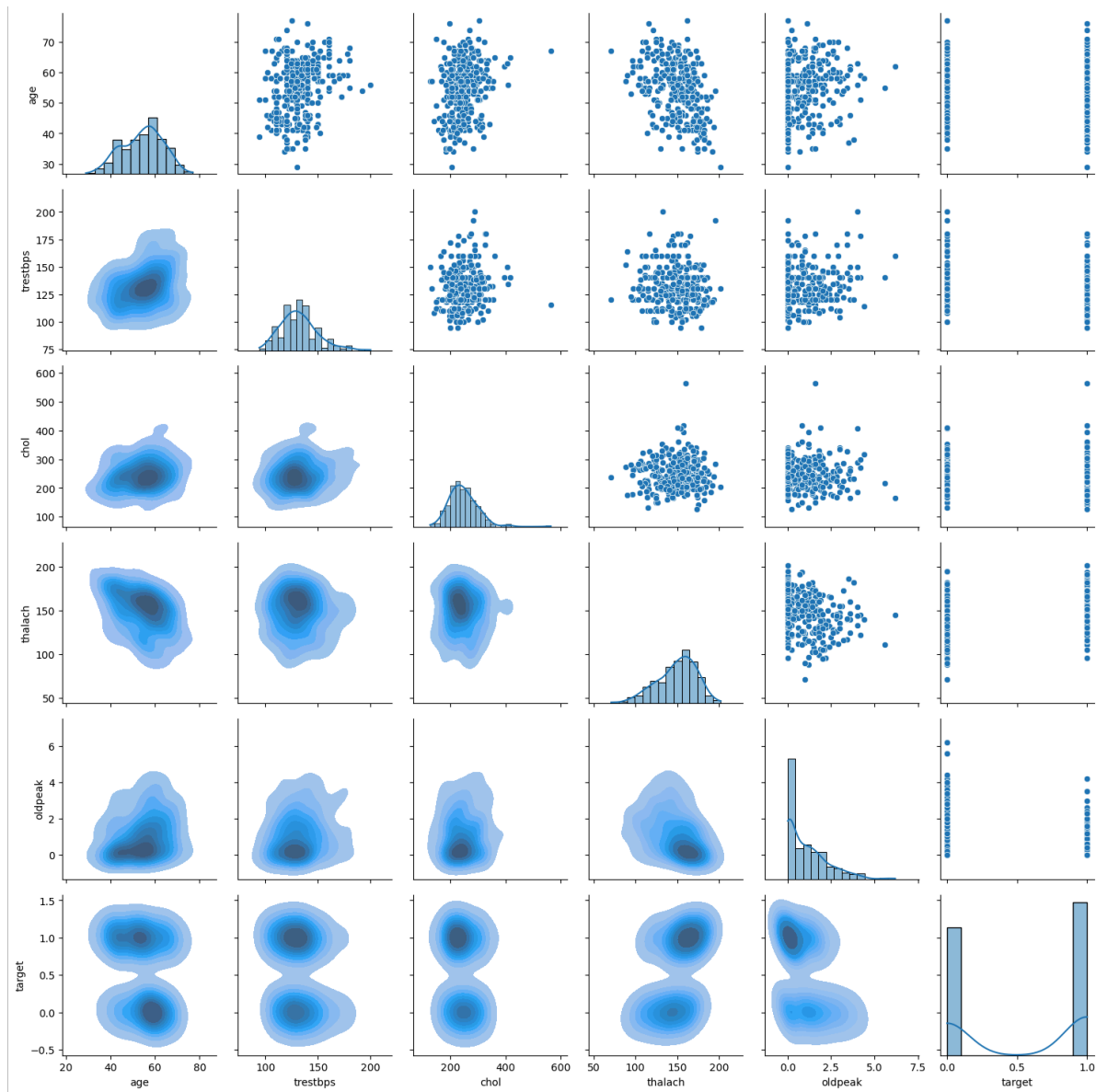


In [101]:

```
g=sns.PairGrid(heart_data[num_var])
g.map_upper(sns.scatterplot)
g.map_lower(sns.kdeplot,fill=True)
g.map_diag(sns.histplot,kde=True)
```

Out[101]:

```
<seaborn.axisgrid.PairGrid at 0x19ec48c8770>
```



In [103]:

```
heart_data['age'].nunique()
```

Out[103]:

41

In [105]:

```
heart_data['age'].describe()
```

Out[105]:

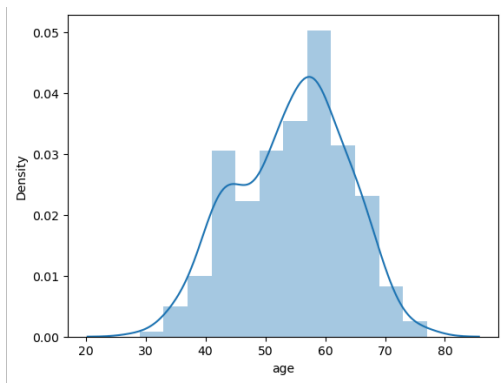
```
count    303.000000
mean      54.366337
std        9.082101
min       29.000000
25%       47.500000
50%       55.000000
75%       61.000000
max       77.000000
Name: age, dtype: float64
```

In [107]:

```
sns.distplot(heart_data['age'])
```

Out[107]:

```
<Axes: xlabel='age', ylabel='Density'>
```

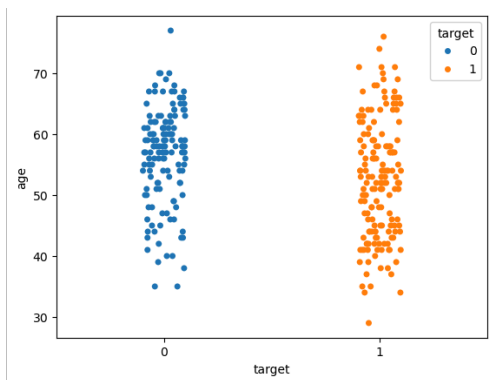


In [109]:

```
sns.stripplot(data=heart_data,x='target',y='age',hue='target')
```

Out[109]:

<Axes: xlabel='target', ylabel='age'>

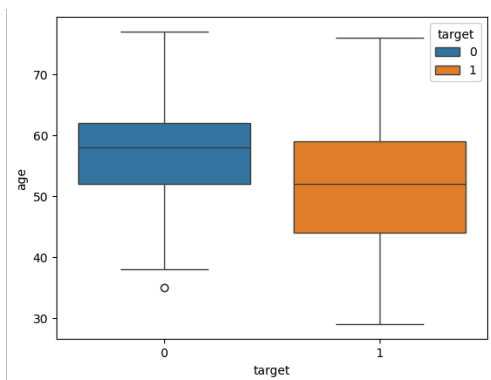


In [112]:

```
sns.boxplot(data=heart_data,x='target',y='age',hue='target')
```

Out[112]:

<Axes: xlabel='target', ylabel='age'>

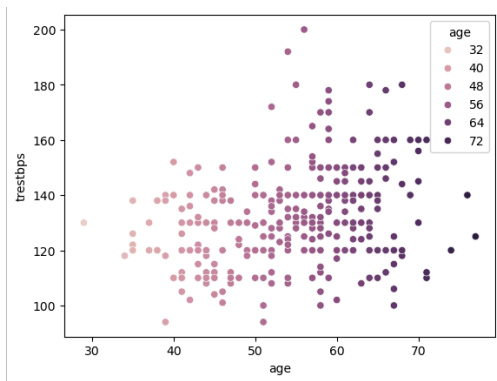


In [114]:

```
sns.scatterplot(data=heart_data,x='age',y='trestbps',hue='age')
```

Out[114]:

<Axes: xlabel='age', ylabel='trestbps'>

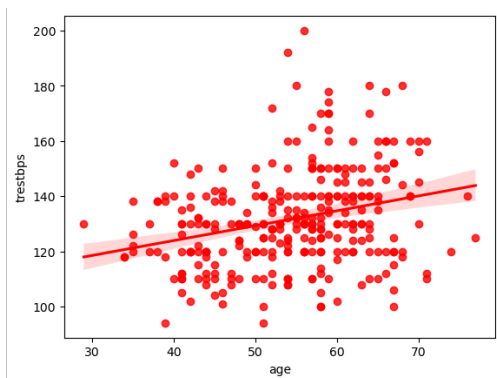


In [116]:

```
sns.regplot(data=heart_data,x='age',y='trestbps',color='r')
```

Out[116]:

<Axes: xlabel='age', ylabel='trestbps'>

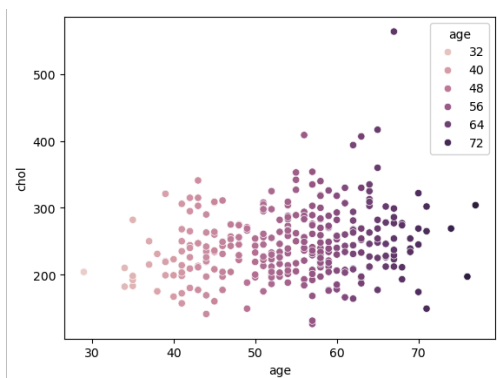


In [118]:

```
sns.scatterplot(data=heart_data,x='age',y='chol',hue='age')
```

Out[118]:

<Axes: xlabel='age', ylabel='chol'>

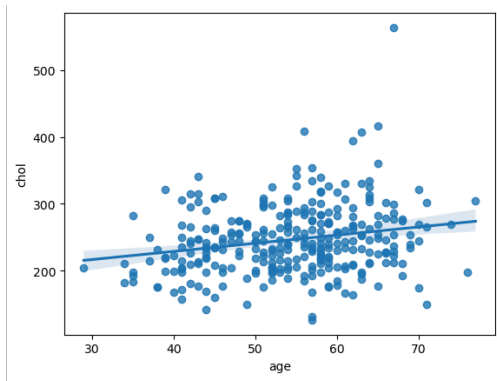


In [120]:

```
sns.regplot(data=heart_data,x='age',y='chol')
```

Out[120]:

<Axes: xlabel='age', ylabel='chol'>

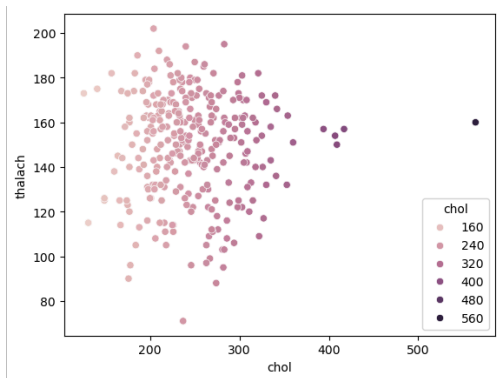


In [122]:

```
sns.scatterplot(data=heart_data,x='chol',y='thalach',hue='chol')
```

Out[122]:

<Axes: xlabel='chol', ylabel='thalach'>

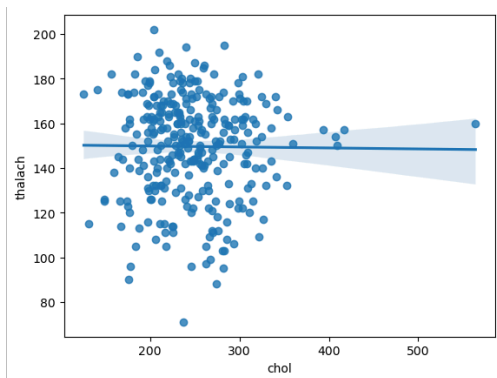


In [124]:

```
sns.regplot(data=heart_data,x='chol',y='thalach')
```

Out[124]:

<Axes: xlabel='chol', ylabel='thalach'>



In [126]:

```
heart_data.isnull().sum()
```

Out[126]:

```
age      0
sex      0
cp       0
trestbps 0
chol     0
fbs      0
restecg  0
thalach  0
exang    0
oldpeak  0
slope    0
ca       0
thal     0
target   0
dtype: int64
```

In [128]:

```
heart_data.isnull().any()
```

Out[128]:

```

age      False
sex      False
cp       False
trestbps False
chol     False
fbs      False
restecg  False
thalach  False
exang    False
oldpeak  False
slope    False
ca       False
thal     False
target   False
dtype: bool

```

In [130]:

```
heart_data.isnull().sum().sum()
```

Out[130]:

```
0
```

In [132]:

```
heart_data.isnull().any().any()
```

Out[132]:

```
False
```

In [134]:

```
heart_data.isnull().mean()
```

Out[134]:

```

age      0.0
sex      0.0
cp       0.0
trestbps 0.0
chol     0.0
fbs      0.0
restecg  0.0
thalach  0.0
exang    0.0
oldpeak  0.0
slope    0.0
ca       0.0
thal     0.0
target   0.0
dtype: float64

```

In [136]:

```
heart_data.isnull().values.any()
```

Out[136]:

```
False
```

In [139]:

```
heart_data.describe()
```

Out[139]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000

In [141]:

```
heart_data['age'].describe()
```

Out[141]:

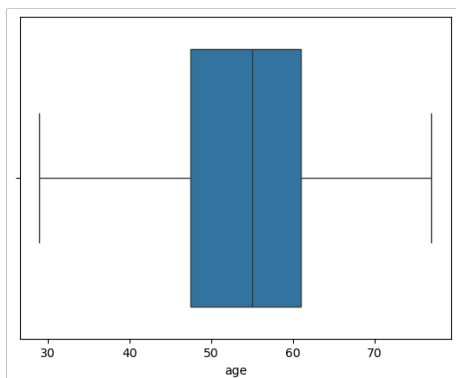
```
count    303.000000
mean      54.366337
std        9.082101
min       29.000000
25%       47.500000
50%       55.000000
75%       61.000000
max       77.000000
Name: age, dtype: float64
```

In [143]:

```
sns.boxplot(x=heart_data['age'])
```

Out[143]:

<Axes: xlabel='age'>



In [145]:

```
heart_data['trestbps'].describe()
```

Out[145]:

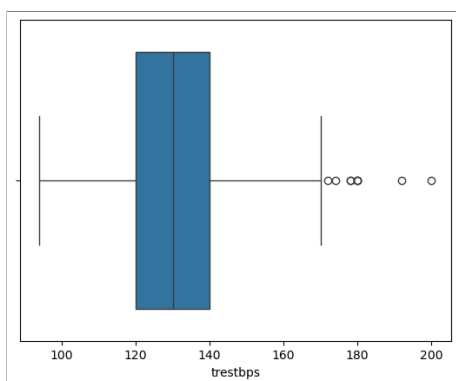
```
count    303.000000
mean     131.623762
std       17.538143
min       94.000000
25%      120.000000
50%      130.000000
75%      140.000000
max      200.000000
Name: trestbps, dtype: float64
```

In [147]:

```
sns.boxplot(x=heart_data['trestbps'])
```

Out[147]:

<Axes: xlabel='trestbps'>



In [149]:

```
heart_data['chol'].describe()
```

Out[149]:

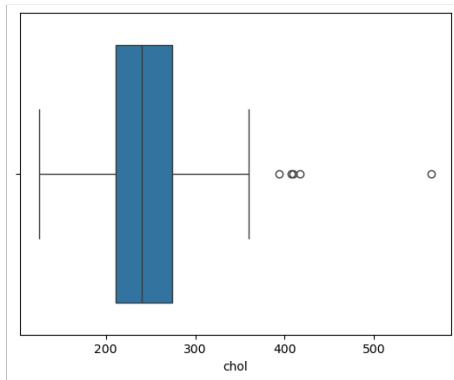
```
count    303.000000
mean     246.264026
std       51.830751
min      126.000000
25%      211.000000
50%      240.000000
75%      274.500000
max      564.000000
Name: chol, dtype: float64
```

In [151]:

```
sns.boxplot(x=heart_data['chol'])
```

Out[151]:

<Axes: xlabel='chol'>



In [153]:

```
heart_data['thalach'].describe()
```

Out[153]:

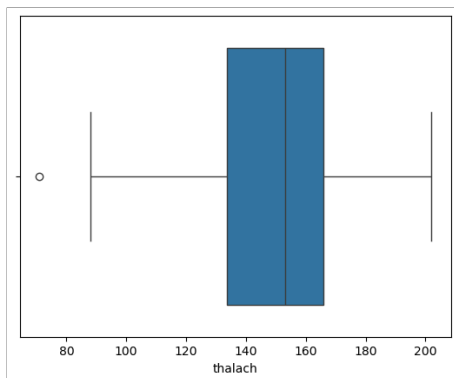
```
count    303.000000
mean     149.646865
std       22.905161
min       71.000000
25%      133.500000
50%      153.000000
75%      166.000000
max       202.000000
Name: thalach, dtype: float64
```

In [155]:

```
sns.boxplot(x=heart_data['thalach'])
```

Out[155]:

<Axes: xlabel='thalach'>



In [157]:

```
heart_data['oldpeak'].describe()
```

Out[157]:

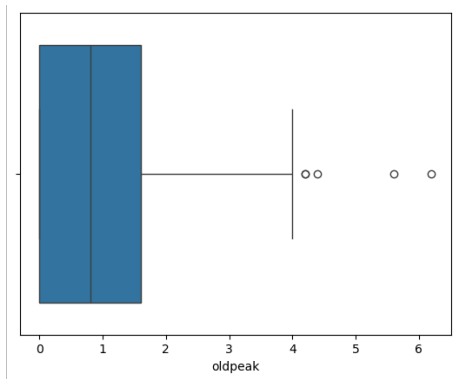
```
count    303.000000
mean       1.039604
std        1.161075
min         0.000000
25%         0.000000
50%         0.800000
75%         1.600000
max         6.200000
Name: oldpeak, dtype: float64
```

In [159]:

```
sns.boxplot(x=heart_data['oldpeak'])
```

Out[159]:

<Axes: xlabel='oldpeak'>



In [161]:

```
heart_data.columns
```

Out[161]:

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',  
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],  
      dtype='object')
```

In [163]:

```
heart_data['sex'].describe()
```

Out[163]:

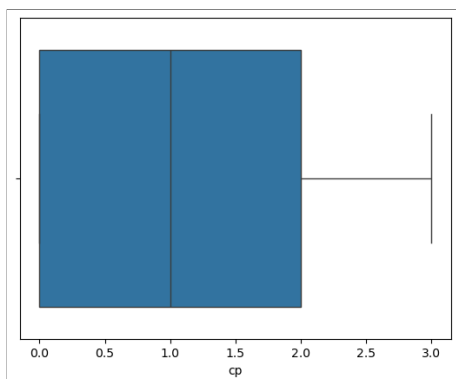
```
count    303.000000  
mean      0.683168  
std       0.466011  
min       0.000000  
25%       0.000000  
50%       1.000000  
75%       1.000000  
max       1.000000  
Name: sex, dtype: float64
```

In [169]:

```
sns.boxplot(x=heart_data['cp'])
```

Out[169]:

```
<Axes: xlabel='cp'>
```



In []: